LECTURE PLAN

NAME OF COURSE	**MCA**
SUBJECT CODE	**KCA204**
SUBJECT NAME	**DATABASE MANAGEMENT SYSTEMS**
BRANCH	**MCA**
SEMESTER	**2ND**
SECTION	**A**
TOTAL NUMBER OF STUDENTS	**26**
NAME OF FACULTY	**JYOTSANA TEWARI**
NUMBER OF LECTURE PROPOSED	**42**

S.No	**Unit No**	**Topic**	**CO**	**No of Lectures Required**	**Actual Date of Completion**	**Suggested Reference**
1	**1**	Overview, Database System vs File System, Database System Concept and Architecture	**1**	**02**		
2		Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure		**03**		
3		Data Modeling Using the Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization		**02**		
4		Aggregation, Reduction of an ER Diagrams to Tables, Extended ER Model, Relationship of Higher Degree		**03**		
No of Lectures Required to complete Unit 1				**10**		
5	**2**	Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Domain Constraints, Relational Algebra, Relational Calculus	**2**	**03**		
6		Tuple and Domain Calculus. Introduction to SQL: Characteristics of SQL, Advantage of SQL. SQL Data Type and Literals. Types of SQL Commands.		**02**		
7		SQL Operators and their Procedure. Tables, Views		**02**		

		and Indexes. Queries and Sub Queries. Aggregate Functions.				
8		Insert, Update and Delete Operations, Joins, Unions, IntersectioInsert, Update and Delete Operations, Joins,		03		
9		Unions, Intersection, Minus, Cursors, Triggers, Procedures in SQL/PL SQL		02		
No of Lectures Required to complete Unit 2				**12**		
10	3	Functional dependencies, normal forms, first, second, third normal forms	3	2		
11		BCNF, inclusion dependence, loss less join decompositions, normalization using FD		2		
12		MVD, and JDs, alternative approaches to database design		2		
No of Lectures Required to complete Unit 3				**06**		
13	4	Transaction System, Testing of Serializability, Serializability of Schedules	4	2		
14		Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery,,		2		
15		Checkpoints, Deadlock Handling. Distributed Database: Distributed Data Storage, Concurrency Control, Directory System		2		
No of Lectures Required to complete Unit 4				**06**		
16	5	Concurrency Control, Locking Techniques forConcurrency Control, Time Stamping Protocols for Concurrency Control	5	2		
17		Validation Based Protocol, Multiple Granularity, Multi Version Schemes, Recovery with Concurrent Transaction, Case Study of Oracle.		2		
No of Lectures Required to complete Unit 5				**04**		

Text Books & References	
1	Korth, Silbertz, Sudarshan," Database Concepts", McGraw Hill.
2	Date C J, "An Introduction to Database Systems", Addision Wesley.
3	Elmasri, Navathe, "Fundamentals of Database Systems", Addision Wesley.
4	O'Neil, "Databases", Elsevier Pub.
5	Majumdar& Bhattacharya, "Database Management System", McGraw Hill.

Course Outcomes (COs)

At the end of this course students will demonstrate the ability to:		
CO 1	Describe the features of a database system and its application and compare various types of data models.	K1, K2
CO 2	Describe the features of a database system and its application and compare various types of data models.	K1, K2
CO 3	Formulate solution to a query problem using SQL Commands, relational algebra, tuple calculus and domain calculus.	K1, K2
CO 4	Explain the need of normalization and normalize a given relation to the desired normal form.	K1, K2
CO 5	Explain different approaches of transaction processing and concurrency control.	K1, K2

Name of Faculty with Signature: MRS. JYOTSANA TEWARI

Lecture Plan In-charge: MR. PRAVEEN KUMAR TRIPATHI

Head of Department: MR. PRASHANT SRIVASTAVA

Dean Academics: DR. KAMAL PRAKASH PANDEY